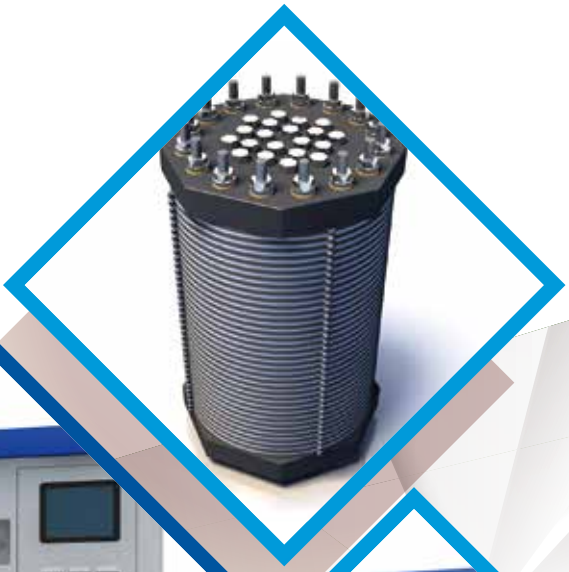
Providing Inno-creative Solutions Worldwide

2023 ENG VER.1

H2Gen®

PEM TYPE Hydrogen Generator



Elchemtech Co., Ltd.

H2Gen ®

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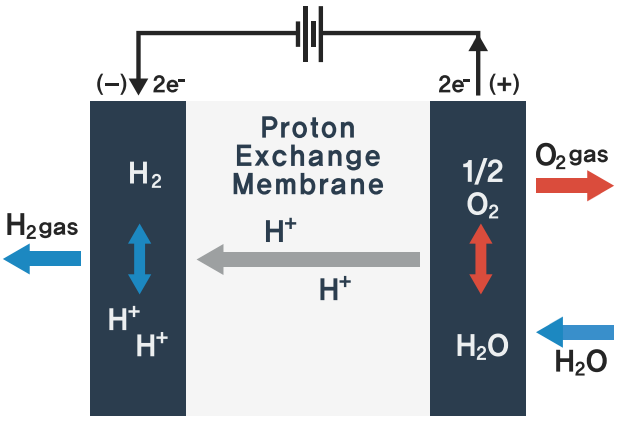
Suitable way to produce Hydrogen

H2Gen® is a PEM type on–stie hydrogen generator

Unlike conventional methods, H2Gen® requires pure water and electricity only.
The byproduct of the system is water and O_2, and DOES NOT produce greenhouse gases like CO_2
H2Gen® is the right system to meet the fluctuation of renewable energy.

H2Gen ® Basic Principle



PEM = Proton Exchange Membrane

① Water is supplied to the Anode, and when electricity is supplied, water is split into the following

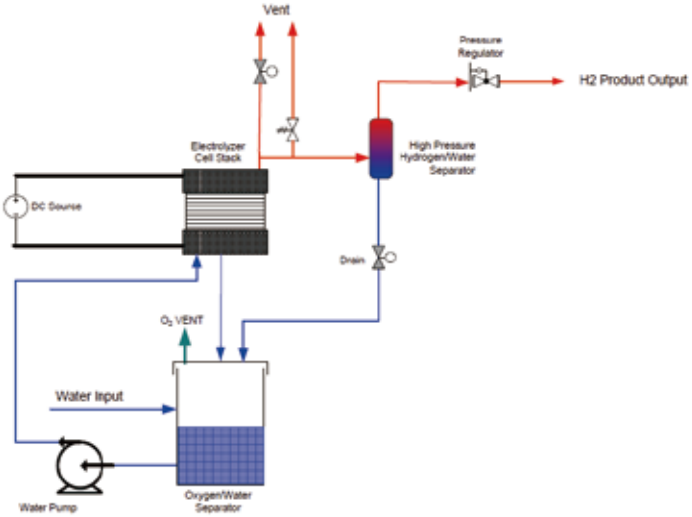
Anode : $2H_2O \rightarrow 4H^+ + 4e^- + O2$

② H^+ ion goes across the PEM to the cathode

③ At Cathode, H^+ ion and electron meets to produce hydrogen

Cathode : $4H^+ + 4e^- \rightarrow 2H_2$

H2Gen ® Characteristic



High Current Density Operation

- Low Capital Cost
- Dynamic Operation

Pressurization of Hydrogen

- Pressurization without compressor (MAX 350bar)

Differential Pressure System

- Pressurization of hydrogen-side
- Simple BOP

H2Gen ® Advantages

Performance

- High Efficiency
- Connection to renewable energy possbile
- Hydrogen compression upto 40bar
- High Purity possible(99.9995%)
- Usage of oxygen possible(Optional)

Economivs

- Easy Maintenance
- Low Management cost

Installation

- Small footprint
- No need for other chemicals

Eco-Friendly

- Low Carbon footprint
- Byproduct: Oxygen, Water

Safety

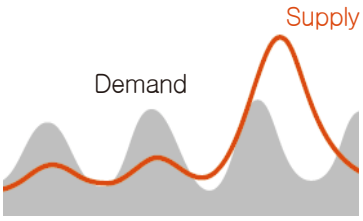
- No Need for caustics
- Oxygen line separation
- Automatic operation

		Lab	Industrial	Refueling	Energy Storage
Value	Safety	V	V	V	V
	Price	V	V	V	V
	Credibility	V	V	V	V
	Compactness	V	V	V	V
	Convenience	V	V	V	V
	CO2 Footprint		V		
	Utilization	V	V		

H2Gen ®

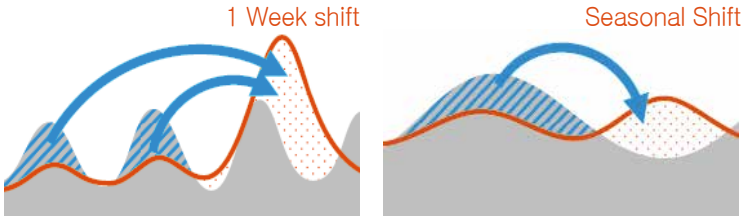
Hydrogen Economy - Eco-Friendly Future

Electricity production of PV



Missmatch of
Supply and Demand

Hydrogen Usage

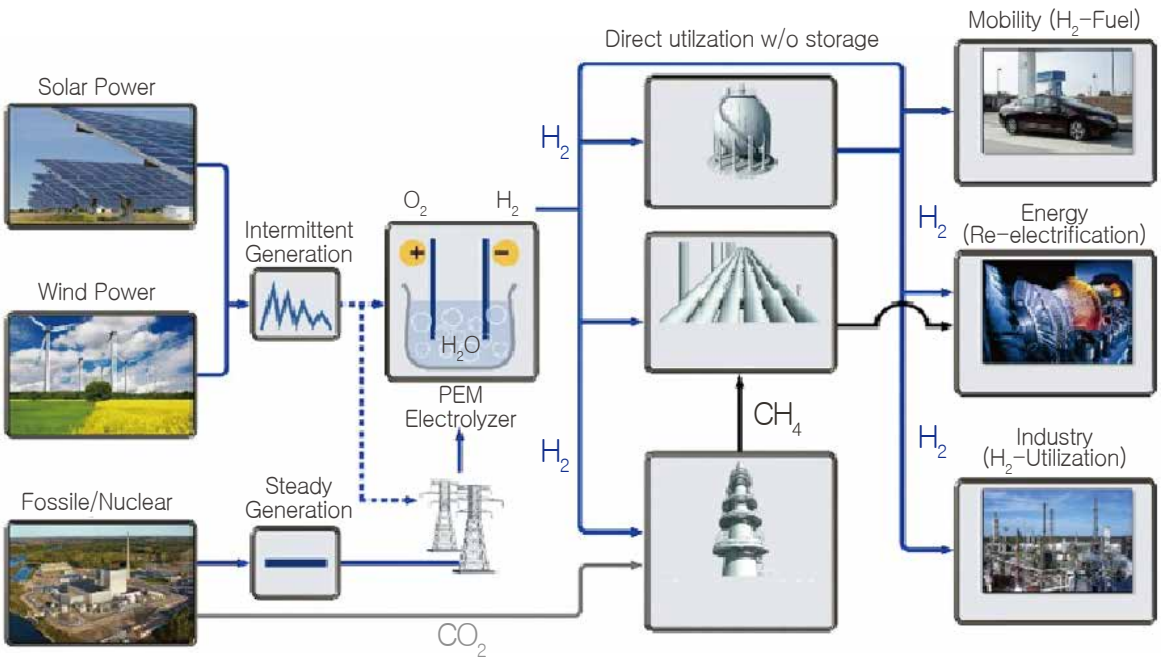


Energy Storage

Hydrogen Refueling

Industrial

Laboratory



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H2Gen ® SPECIFICATION

H2GEN®									
DESCRIPTION		PEM Type Hydrogen Generator Proton Exchange Memebrane							
Product		H2Gen 5	H2Gen 10	H2Gen 20	H2Gen 100	H2Gen 200	H2Gen 300	H2Gen 1000	H2Gen 2000
Production (Nm^3/hr, 0℃)		0.5 Nm^3	1.0 Nm^3	2.0 Nm^3	10 Nm^3	20 Nm^3	30 Nm^3	100 Nm^3	200 Nm^3
Power Consumption at stack		4.8 kWh/Nm^3 (Nominal, BOL)							
Delivery Pressure	Basic	7.2 bar							
	Option 1	30 bar							
	Option 2	350 bar (Max 0.2Nm^3/hr)							
Hydrogen Purity	Basic	99.95%							
	Option 1	99.995%							
	Option 2	99.9995%							
	Reference	• N2 〈 2 ppm, O2 〈 2 ppm. • Other Gas undetected							
DI Water Requirement	Required Preferred	ASTM Type II Deionizer Water 〈 1 micro Siemen/cm (〉 1 MegOhm-cm) ASTM Type I Deionizer Water 〈 0.1 micro Siemen/cm (〉10 MegOhm-cm)							
Electricity Specification		380 to 480 VAC , 3 phase , 50 or 60 Hz							
Operation Condition		Indoors, Non- hazardous / Non-classified environment							
Storage/Transport Temperature		5 ~ 60℃							
Operating Temperature		55℃							
Ventilation		Ventilation must be provided from non Hazardous area, at a rate in accordance with IEC60079-10							
Options		• DI water system • Dew Point Monitoring • Regenerative Energy connection • Containerization							

※ For other production rate, please consult with Elchemtech Personnel

H2Gen ®

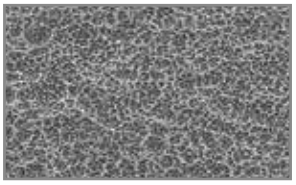
Why Elchemtech?

Name	Elchemtech Co., Ltd
Established	2002. 04
CEO	Dr. Sangbong Moon
Main Business	전기화학을 기반으로 한 살균기 수소/산소 발생기, BWT 설비
Headquater	Gangseogu, Seoul
Factory	Gangseogu, Seoul
Pilot Station	Ganhwa, Incheon
Employee	35
Patents	46
Certificate	Innobiz I NET 제1147호 ISO 9001:2008 I ISO 14001:2004

Material Development → System Production

Catalyst

Electrode



MEA

Stack

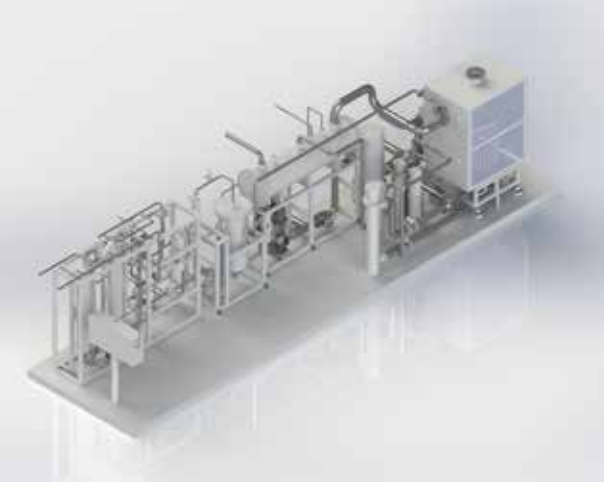
BWT System

H2Gen

HyGenOn

R&D

More than a decade of research on PEM



21c New Fontier Project
(2003~2013)

- High Efficiency PEM water electrolyzer

Nano 2020 Project
(2015~2018)

- Active Area: 3,000cm2

Development of MW size electrolyzer
(2018~2021)

- 1MW size (200Nm^3/hr) PEM type electrolyzer developemnt

Power to Gas Project
(2019~2022)

References

- New Excellent Technology
- 2018 Top 10 Nano Technology Award
- 2018.09 Minister of Finance visiting Elchemtech
- 2018.10 Minister of MOTIE Award
- 2019.01 Reception of President of RoK
- 2019.05 Minister of Environment Award
- 2020.12 Delivery of 20Nm^3/h H_2 Generator
- 2021.12 Delivery of 200Nm^3/h H_2 Generator
- 2022.01 Delivery of 10Nm^3/h H_2 Generator
- 2022.03 Delivery of 10Nm^3/h H_2 Generator
- 2022.07 Purchase Order for 20Nm^3/h H_2 Generator
- 2022.10 Purchase Order for 2MW PEM Stack
- 2023.01 Purchase Order for 300kW H_2 Generator

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